

Visual Spatial Reasoning

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Abstract

Spatial relations are a basic part of human cognition. However, they are expressed in natural language in a variety of ways, and previous work has suggested that current vision-and-language models (VLMs) struggle to capture relational information. In this paper, we present Visual Spatial Reasoning (VSR), a dataset containing more than 10k natural text-image pairs with 66 types of spatial relations in English (such as: under, in front of, facing). While using a seemingly simple annotation format, we show how the dataset includes challenging linguistic phenomena, such as varying reference frames. We demonstrate a large gap between human and model performance: the human ceiling is above 95%, while state-of-the-art models only achieve around 70%. We observe that VLMs’ by-relation performances have little correlation with the number of training examples and the tested models are in general incapable of recognising relations concerning the orientations of objects.¹

1 Introduction

Multimodal NLP research has developed rapidly in recent years, with substantial performance gains on tasks such as visual question answering (VQA) (Antol et al., 2015; Johnson et al., 2017; Goyal et al., 2017; Hudson and Manning, 2019; Zellers et al., 2019), vision-language reasoning or entailment (Suhr et al., 2017, 2019; Xie et al., 2019; Liu et al., 2021), and referring expression comprehension (Yu et al., 2016; Liu et al., 2019). Existing benchmarks, such as NLVR2 (Suhr et al., 2019) and VQA (Goyal et al., 2017), define generic paradigms for testing vision-language models (VLMs). However, as we further discuss in §2, these benchmarks are not ideal for probing VLMs as they typically

conflate multiple sources of error and do not allow controlled analysis on specific linguistic or cognitive properties, making it difficult to categorise and fully understand the model failures. In particular, spatial reasoning has been found to be particularly challenging for current models, and much more challenging than capturing properties of individual entities (Kuhnle et al., 2018; Cirik et al., 2018; Akula et al., 2020), even for state-of-the-art models such as CLIP (Radford et al., 2021; Subramanian et al., 2022).

Another line of work generates synthetic datasets in a controlled manner to target specific relations and properties when testing VLMs, e.g., CLEVR (Liu et al., 2019) and ShapeWorld (Kuhnle and Copestake, 2018). However, synthetic datasets may accidentally overlook challenges (such as orientations of objects which we will discuss in §5), and using natural images allows us to explore a wider range of language use.

To address the lack of probing evaluation benchmarks in this field, we present VSR (Visual Spatial Reasoning), a controlled dataset that explicitly tests VLMs for spatial reasoning. We choose spatial reasoning as the focus because it is one of the most fundamental capabilities for both humans and VLMs. Such relations are crucial to how humans organise their mental space and make sense of the physical world, and therefore fundamental for a grounded semantic model (Talmy, 1983).

The VSR dataset contains natural image-text pairs in English, with the data collection process explained in §3. Each example in the dataset consists of an image and a natural language description which states a spatial relation of two objects presented in the image (two examples are shown in Fig. 1 and Fig. 2). A VLM needs to classify the image-caption pair as either true or false, indicating whether the caption is correctly describing the spatial relation. The dataset covers 66 spatial relations and has >10k data points, using 6,940 images from

¹Data and code: github.com/cambridgeltl/visual-spatial-reasoning.

MS COCO (Lin et al., 2014).

Situating one object in relation to another requires a *frame of reference*: a system of coordinates against which the objects can be placed. Drawing on detailed studies of more than forty typologically diverse languages, Levinson (2003) concludes that the diversity can be reduced to three major types: intrinsic, relative, and absolute. An intrinsic frame is centred on an object, e.g., *behind the chair*, meaning at the side with the backrest. A relative frame is centred on a viewer, e.g., *behind the chair*, meaning further away from someone’s perspective. An absolute frame uses fixed coordinates, e.g., *north of the chair*, using cardinal directions. In English, absolute frames are rarely used when describing relations on a small scale, and they do not appear in our dataset. However, intrinsic and relative frames are widely used, and present an important source of variation. We discuss the impact on data collection in §3.2, and analyse the collected data in §4.

We test four popular VLMs, i.e., VisualBERT (Li et al., 2019), LXMERT (Tan and Bansal, 2019), ViLT (Kim et al., 2021), and CLIP (Radford et al., 2021) on VSR, with results given in §5. While the human ceiling is above 95%, all four models struggle to reach 70% accuracy. We conduct comprehensive analysis on the failures of the investigated VLMs and highlight that (1) positional encodings are extremely important for the VSR task; (2) models’ by-relation performance barely correlates with the number of training examples; (3) in fact, several spatial relations that concern orientation of objects are especially challenging for current VLMs; (4) VLMs have extremely poor generalisation on unseen concepts.

2 Related Work

2.1 Comparison with synthetic datasets

Synthetic language-vision reasoning datasets, e.g., SHAPES (Andreas et al., 2016), CLEVR (Liu et al., 2019), NLVR (Suhr et al., 2017), and ShapeWorld (Kuhnle and Copestake, 2018), enable full control of dataset generation and could potentially benefit probing of spatial reasoning capability of VLMs. They share a similar goal to us, to diagnose and pinpoint weaknesses in VLMs. However, synthetic datasets necessarily simplify the problem as they have inherently bounded expressivity. In CLEVR, objects can only be spatially related via four relationships: “left”, “right”, “behind”, and “in front of” while VSR covers 66 relations.



Figure 1: Caption: *The potted plant is at the right side of the bench.* Label: True.



Figure 2: Caption: *The cow is ahead of the person.* Label: False.

Synthetic data does not always accurately reflect the challenges of reasoning in the real world. For example, objects like spheres, which often appear in synthetic datasets, do not have orientations. In real images, orientations matter and human language use depends on that. Furthermore, synthetic images do not take the scene as a context into account. The interpretation of object relations can depend on such scenes (e.g., the degree of *closeness* can vary in open space and indoor scenes).

Last but not least, the vast majority of spatial relationships cannot be determined by rules. Even for the seemingly simple relationships like “left/right of”, the determination of two objects’ spatial relationships can depend on the observer’s viewpoint, whether the object has a *front*, if so, what are their orientations, etc.

2.2 Spatial relations in existing vision-language datasets

Several existing vision-language datasets with natural images also contain spatial relations (e.g., NLVR2, COCO, and VQA datasets). Suhr et al. (2019) summarise that there are 9 prevalent linguistic phenomena/challenges in NLVR2 (Suhr et al., 2019) such as coreference, existential quantifiers, hard cardinality, spatial relations, etc., and 4 in

VQA datasets (Antol et al., 2015; Hudson and Manning, 2019). However, the different challenges are entangled in these datasets. Sentences contain complex lexical and syntactic information and can thus conflate different sources of error, making it hard to identify the exact challenge and preventing categorised analysis. Yatskar et al. (2016) extract 6 types of visual spatial relations directly from MS COCO images with annotated bounding boxes. But rule-based automatic extraction can be restrictive as most relations are complex and cannot be identified relying on bounding boxes. Recently, Rösch and Libovický (2022) extract captions that contain 28 positional keywords from MS COCO and swap the keywords with their antonyms to construct a challenging probing dataset. However, the COCO captions also have the error-conflation problem. Also, the number of examples and types of relations are restricted by COCO captions.

Visual Genome (Krishna et al., 2017) also contains annotations of objects’ relations including spatial relations. However, it is only a collection of true statements and contains no negative ones, so cannot be framed as a binary classification task. It is non-trivial to automatically construct negative examples since multiple relations can be plausible for a pair of object in a given image. Relation classifiers are harder to learn than object classifiers on this dataset (Liu and Emerson, 2022).

Parcalabescu et al. (2022) propose a benchmark called VALSE for testing VLMs’ capabilities on various linguistic phenomena. VALSE has a sub-set focusing on “relations” between objects. It uses texts modified from COCO’s original captions. However, it is a zero-shot benchmark without training set, containing just 535 data points. So, it is not ideal for large-scale probing on a wide spectrum of spatial relations.

2.3 Spatial reasoning without grounding

There has also been interest in probing models’ spatial reasoning capability without visual input. For example, Collell et al. (2018); Mirzaee et al. (2021); Liu et al. (2022) probe pretrained text-only models or VLMs’ spatial reasoning capabilities with text-only questions. However, a text-only dataset cannot evaluate how a model relates language to grounded spatial information. In contrast, VSR focuses on the joint understanding of vision and language input.

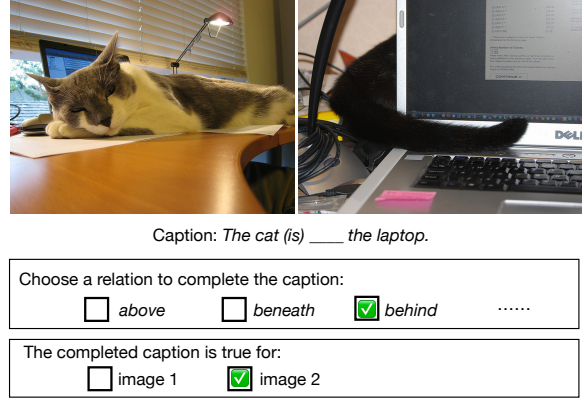


Figure 3: An annotation example of concepts “cat” & “laptop” in contrastive caption generation. The example generates two data points for our dataset: one “True” instance when the completed caption is paired with image 2 (right) and one “False” instance when paired with image 1 (left).

2.4 Spatial reasoning as a sub-component

Last but not least, some vision-language tasks and models require spatial reasoning as a sub-component. For example, Lei et al. (2020) propose TVQA+, a spatio-temporal video QA dataset containing bounding boxes for objects referred in the questions. Models then need to simultaneously conduct QA while detecting the correct object of interest. Christie et al. (2016) propose a method for simultaneous image segmentation and prepositional phrase attachment resolution. Models have to reason about objects’ spatial relations in the visual scene to determine the assignment of prepositional phrases. However, if spatial reasoning is only a sub-component of a task, error analysis becomes more difficult. In contrast, VSR provides a focused evaluation of spatial relations, which are particularly challenging for current models.

3 Dataset Creation

In this section we detail how VSR is constructed. The data collection process can generally be split into two phases (1) contrastive caption generation (§3.1) and (2) second-round validation (§3.2). We then discuss annotator hiring & payment (§3.3), dataset splits (§3.4), and the human ceiling & agreement of VSR (§3.5).

3.1 Contrastive Template-based Caption Generation (Fig. 3)

In order to highlight spatial relations and avoid annotators frequently choosing trivial relations (such